

Deep Learning — Week 1 Notes

NPTEL · IIT Ropar | From Neurons to Perceptrons

Covers: Biological & Artificial Neurons · McCulloch-Pitts Neuron · Thresholding Logic · Perceptron · Perceptron Learning Algorithm & Convergence · Linear Separability · XOR & Networks of Perceptrons

1 · Biological → Artificial Neuron

The **artificial neuron** is the most fundamental unit of a deep neural network; its inspiration comes from the brain.

- **Dendrite** → receives signals from other neurons
- **Synapse** → point of connection to other neurons
- **Soma** → processes the information
- **Axon** → transmits the output of the neuron

A neuron takes inputs, **aggregates** them, and **decides** whether to "fire". The brain has $\sim 10^{11}$ neurons in a massively parallel, **hierarchical** network (e.g. visual cortex $V1 \rightarrow V2 \rightarrow \dots \rightarrow AIT$: simple forms $\rightarrow$ high-level objects).

inputs $x_i \rightarrow$ aggregate $g(x) \rightarrow$ decide $f \rightarrow$ output y

2 · McCulloch-Pitts (MP) Neuron — 1943

A highly simplified model. Inputs are **boolean** (0/1); output $y \in \{0,1\}$. The function **g** **aggregates** inputs and **f** **thresholds** the sum. This is called **Thresholding Logic**.

$$g(x) = \sum_{i=1}^n x_i$$
$$y = f(g(x)) = 1 \text{ if } g(x) \geq \theta, \text{ else } 0$$

θ = **thresholding parameter**. Inputs may be **excitatory** or **inhibitory**: if **any inhibitory input = 1, output is forced to 0** (shown as a circle at the line's end).

Boolean functions via one MP unit ($n = 2$):

Function	Threshold θ
AND	$\theta = 3$ ($n=3$ in slide)
AND (2-input)	$\theta = 2$
OR	$\theta = 1$
NOR	$\theta = 0$
Tautology (always ON)	$\theta = 0$
x_1 AND $\neg x_2$	inhibitory x_2

Rule of thumb: an MP unit fires when "at least θ of the inputs are 1".

3 · Geometric View & Linear Separability

A single MP neuron **splits the input space into two halves** with the line/plane $\sum x_i - \theta = 0$. Points with output 1 lie on one side ($\sum x_i \geq \theta$); output 0 on the other ($\sum x_i < \theta$). With 2 inputs it is a **line**; with 3 inputs it is a **plane**.

Linear separability (boolean): there exists a line/plane such that all inputs producing 1 lie on one side and all inputs producing 0 lie on the other. **A single MP neuron can only represent linearly separable functions.**

4 · Perceptron — Rosenblatt 1958

A more general model than MP. Two key additions:
numerical weights for inputs, and a **mechanism for learning** them. Inputs need not be boolean (real values allowed).

$$y = 1 \text{ if } \sum w_i x_i \geq \theta, \text{ else } 0 \\ \Leftrightarrow \sum_{i=0}^n w_i x_i \geq 0, \text{ with } x_0=1, w_0=-\theta$$

$w_0 = -\theta$ is the bias — it encodes the "prior/prejudice".
Low threshold = fires easily ($\theta=0$); high threshold = selective (θ large).

MP vs Perceptron

- MP: unweighted inputs, hand-set θ , boolean only.
- Perceptron: **weighted** inputs, **learnable** θ , **real-valued** inputs.
- **Both** can implement **only linearly separable** functions with a single unit.

OR via perceptron: one valid solution is

$$w_0=-0.5, w_1=1, w_2=1 \rightarrow \text{boundary } x_1+x_2-0.5=0$$

Only (0,0) gives a negative score; all other inputs are ≥ 0 .

5 · Perceptron Learning Algorithm (PLA)

Used as a **binary classifier**. Assumes data is **linearly separable**; learns w so positive points get $w \cdot x \geq 0$ and negatives get $w \cdot x < 0$.

```
P ← inputs with label 1;   N ← inputs with label 0;
Initialize w randomly;
while !convergence do
  Pick random x ∈ P ∪ N;
  if x ∈ P and w · x < 0 then  w = w + x;
  if x ∈ N and w · x ≥ 0 then  w = w - x;
// converges when all inputs are classified correctly
```

Why it works (geometry)

w is **perpendicular** to the decision line $w \cdot x = 0$ (angle 90° , since $\cos \alpha = w \cdot x / \|w\| \|x\| = 0$). Positive points make angle $< 90^\circ$ with w ; negative points $> 90^\circ$.

For $x \in P$ with $w \cdot x < 0$ (angle too big), set $w_{\text{new}} = w + x$:

$$\cos \alpha_{\text{new}} \propto w \cdot x + x \cdot x > \cos \alpha$$

→ angle shrinks (good). For $x \in N$ we subtract x , so the angle grows. Each update nudges w the right way.

6 · Perceptron Convergence Theorem

Statement: If P and N are finite and **linearly separable**, the PLA makes only a **finite number of corrections** and finds a separating w in finite steps. **Linear separability is the required condition.**

Proof idea: trick — replace each $n \in N$ by $-n$, so we need $w \cdot p \geq 0$ for all p ; normalize $\|p\| = 1$. Let w^* be a solution, β the angle between w^* and w_t . After k corrections:

$$\begin{aligned} \text{numerator} &\geq w^* \cdot w_0 + k\delta \quad (\text{grows } \propto k) \quad ; \quad \|w_t\|^2 \leq \|w_0\|^2 + k \\ \Rightarrow \cos \beta &\geq (w^* \cdot w_0 + k\delta) / \sqrt{(\|w_0\|^2 + k)} \quad \text{grows } \propto \sqrt{k} \end{aligned}$$

But $\cos \beta \leq 1$ always, so **k must be bounded** → only finitely many corrections → the algorithm converges. ($\delta = \min_i w^* \cdot p_i > 0$.)

7 · XOR & Networks of Perceptrons

XOR is NOT linearly separable. Its inequalities contradict (the 4th contradicts 2 & 3), and geometrically the two "1" points (0,1) & (1,0) are on **diagonally opposite corners** — no single line separates them.

x_1	x_2	XOR
0	0	0
0	1	1
1	0	1
1	1	0

Of the 16 boolean functions of 2 inputs, all are linearly separable **except XOR and !XOR**.

Counting: boolean functions of n inputs = $2^{(2^n)}$. Layers: **input** → **hidden** → **output**.

Representation power

A single perceptron can't do XOR, but a **network** can implement **any** boolean function.

Any boolean function of n inputs can be represented exactly by a network with **one hidden layer of 2^n perceptrons** + 1 output perceptron.

Each hidden unit fires for exactly one input pattern; each 2nd-layer weight tunes the output for that pattern → no contradictions.

Catch: hidden units grow exponentially with n .

8 · Formula Cheat-Sheet

MP: $y=1$ if $\sum x_i \geq \theta$

Perceptron: $y=1$ if $\sum w_i x_i \geq 0$

bias $w_0 = -\theta$

OR: $(-0.5, 1, 1)$

boundary: $x_1 + x_2 - 0.5 = 0$

PLA(P): $w \leftarrow w + x$

PLA(N): $w \leftarrow w - x$

update (± 1 labels): $w \leftarrow w + y \cdot x$

boolean fns = $2^{(2^n)}$

hidden units = 2^n

converges $\Leftrightarrow$ linearly separable

9 · Exam Quick-Fire (assignment-style)

Q. Threshold to fire when ≥ 3 of 4 indicators active?

$\theta = 3$. MP unit fires when at least θ inputs are 1.

Q. How many patterns switch the fan ON (≥ 3 of 4)?

$5 = C(4,3) + C(4,4) = 4 + 1$.

Q. Rule "flag if ≥ 1 of 2 conditions" is which gate?

OR — linearly separable, one perceptron works.

Q. Correct OR parameters?

$w_1=1, w_2=1, \mathbf{b=-0.5}$ → only (0,0) is negative.

Q. Score of OR unit for input (0,1)?

$1 \cdot 0 + 1 \cdot 1 - 0.5 = \mathbf{0.5}$ → outputs 1.

Q. "Flag exactly one returned" is which gate?

XOR — not linearly separable (diagonal corners).

Q. Which modified rule IS linearly separable?

"Flag only when both returned" = **AND**.

Q. Perceptron update (labels ± 1) when $y(w \cdot x) \leq 0$?

$w \leftarrow w + y \cdot x$. E.g. $w=(0,0)+1 \cdot (2,1)=(2,1)$.

Q. Condition for the convergence guarantee?

Data must be **linearly separable**.

Q. Vector separating

$P1(2,1,+), P2(1,2,+), P3(-1,-1,-)$

$w=(1,1)$: $P1, P2$ score > 0 , $P3 < 0$.

☆ Remember: single unit $\Rightarrow$ linear separability only · bias = $-\theta$ · network of perceptrons $\Rightarrow$ any boolean function ☆